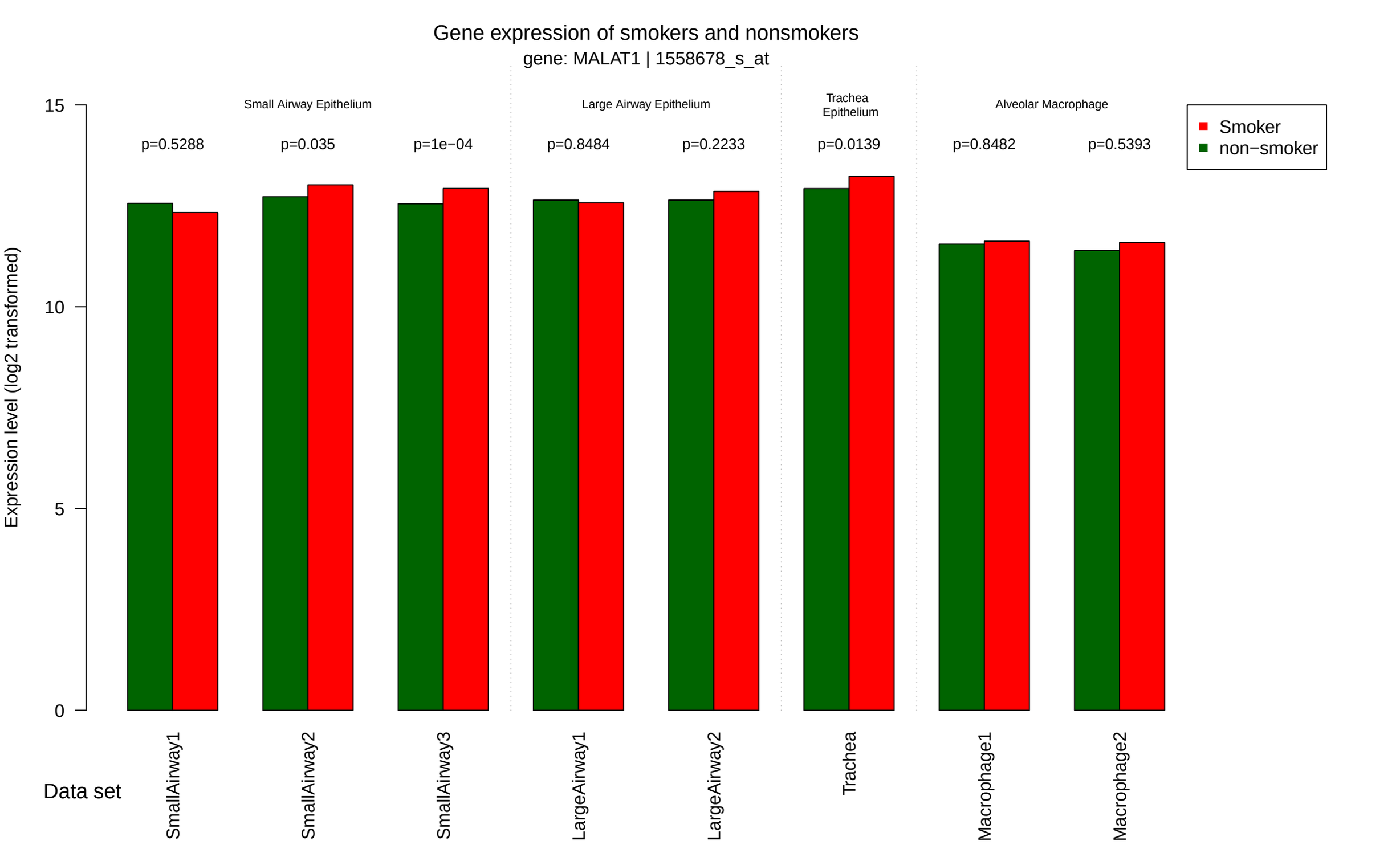
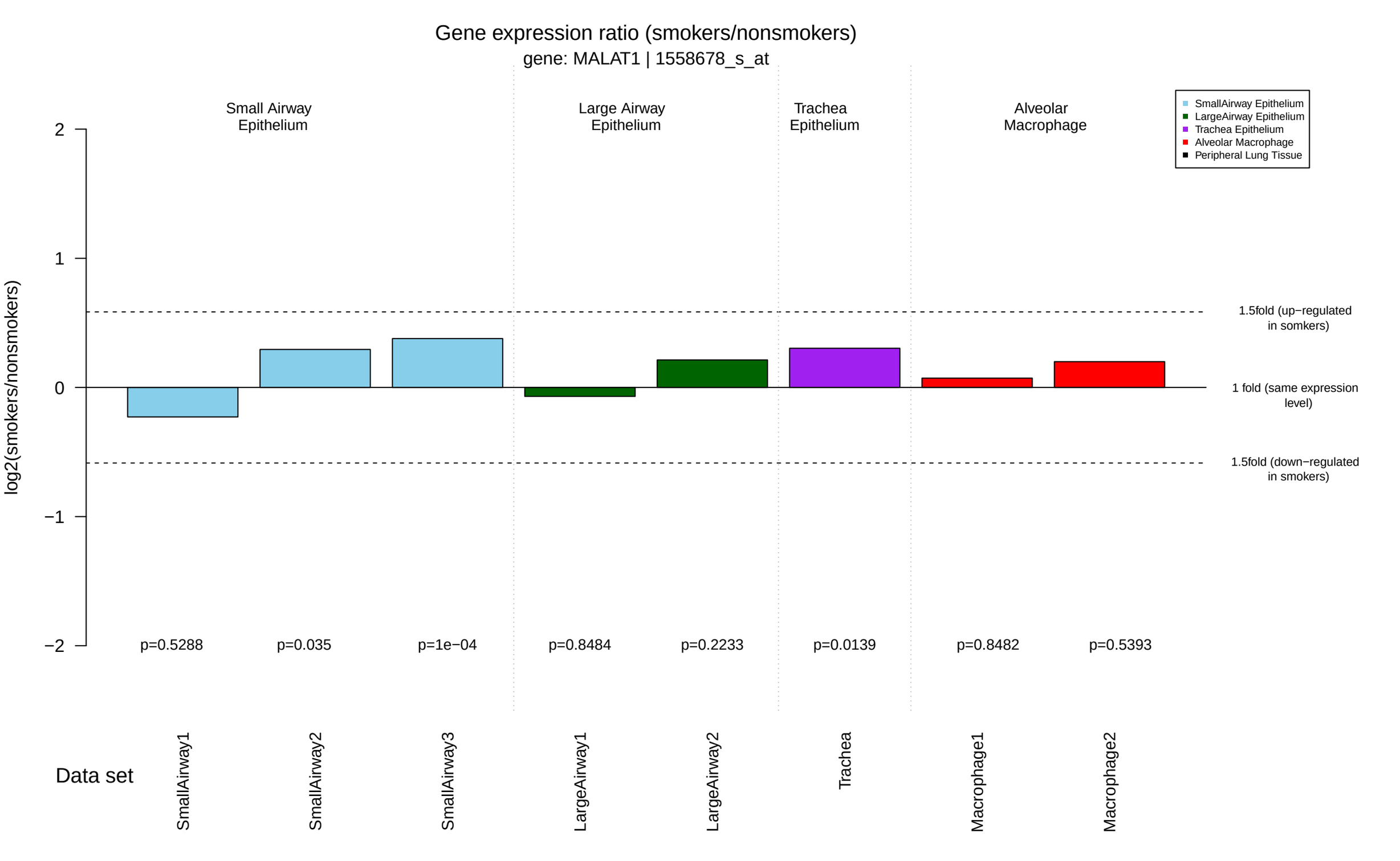
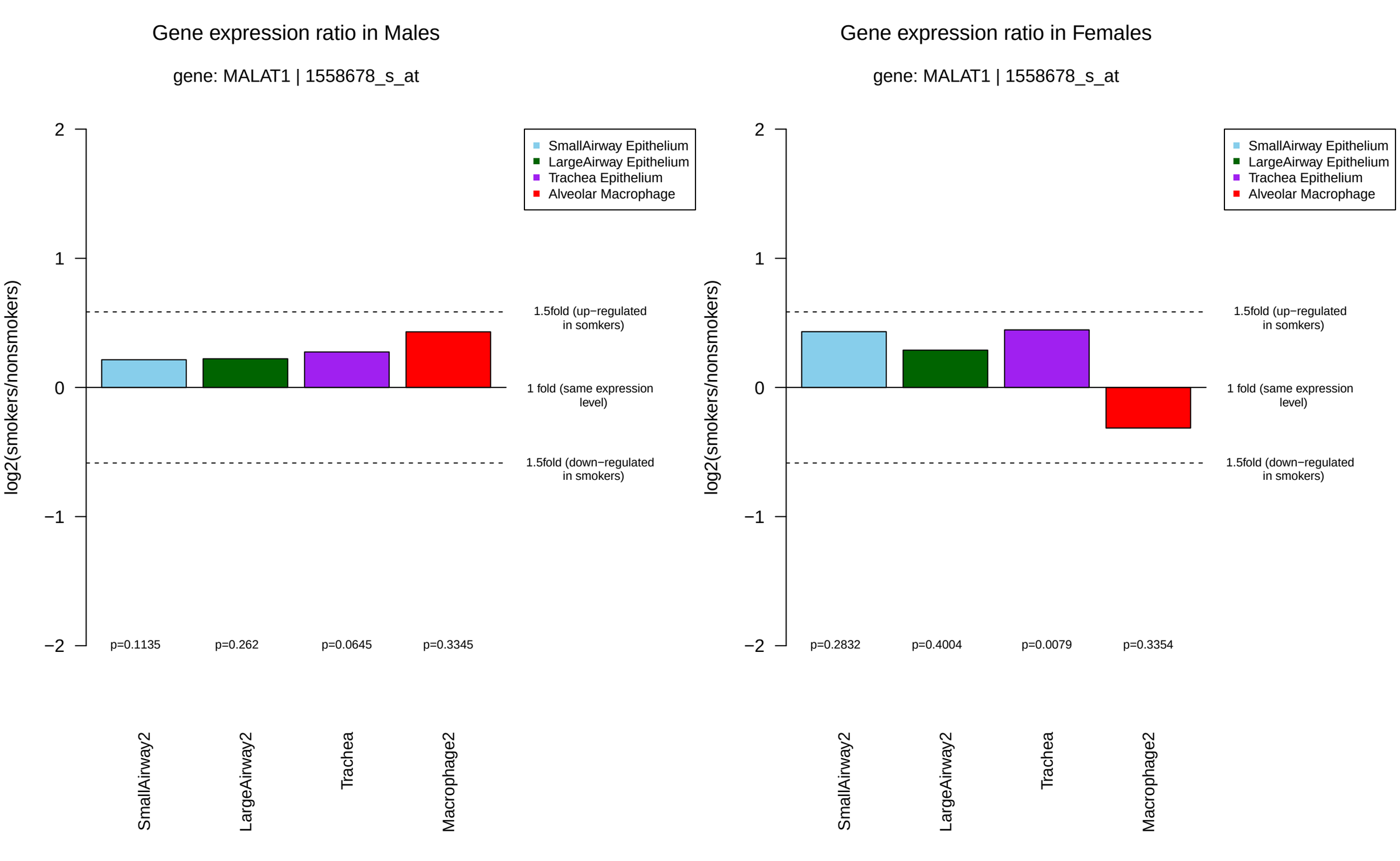
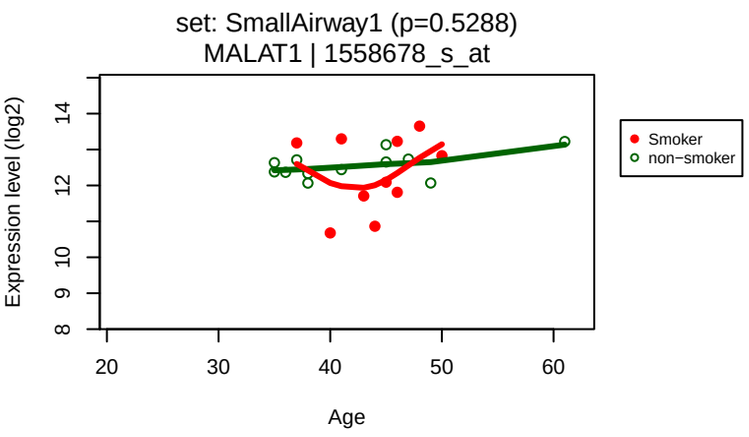
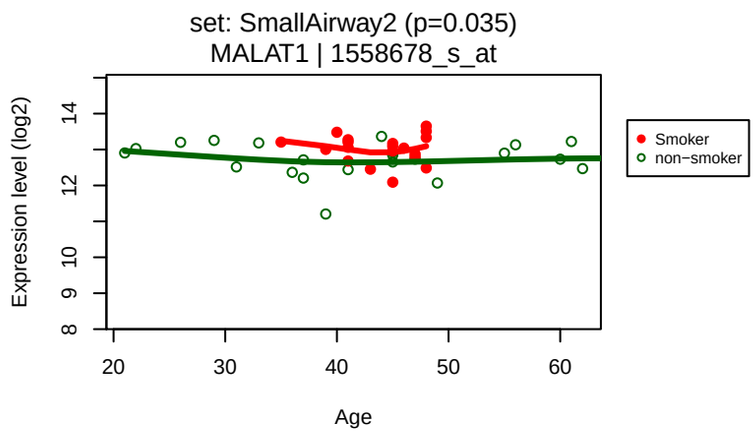
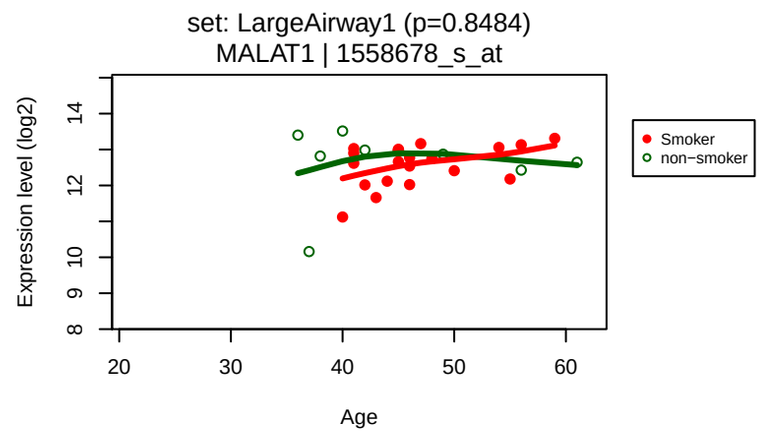

Gene expression of smokers and nonsmokers
gene: MALAT1 | 1558678_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5288
p=0.035
p=1e−04
p=0.8484
p=0.2233
p=0.0139
p=0.8482
p=0.5393
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

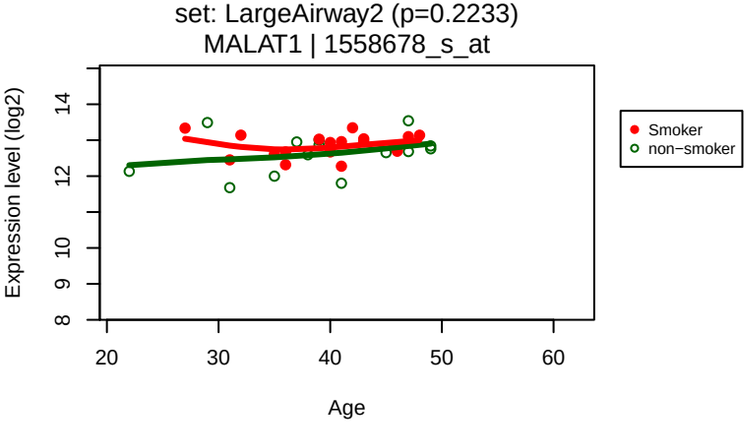

Gene expression ratio (smokers/nonsmokers)
gene: MALAT1 | 1558678_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5288
p=0.035
p=1e−04
p=0.8484
p=0.2233
p=0.0139
p=0.8482
p=0.5393
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

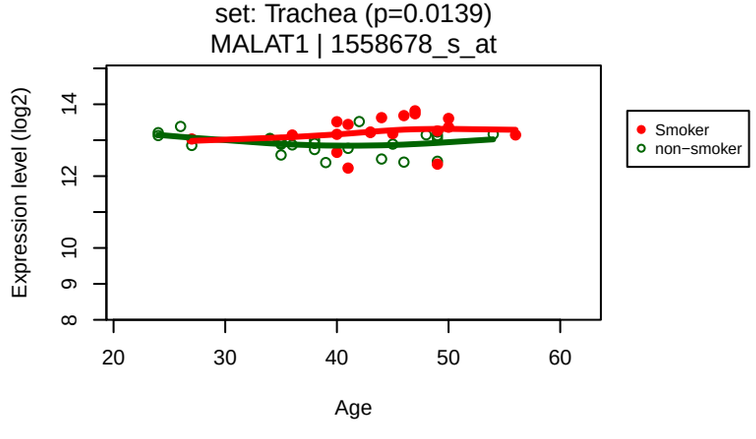

Gene expression ratio in Males
gene: MALAT1 | 1558678_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1135
p=0.262
p=0.0645
p=0.3345
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MALAT1 | 1558678_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2832
p=0.4004
p=0.0079
p=0.3354
SmallAirway2
LargeAirway2
Trachea
Macrophage2

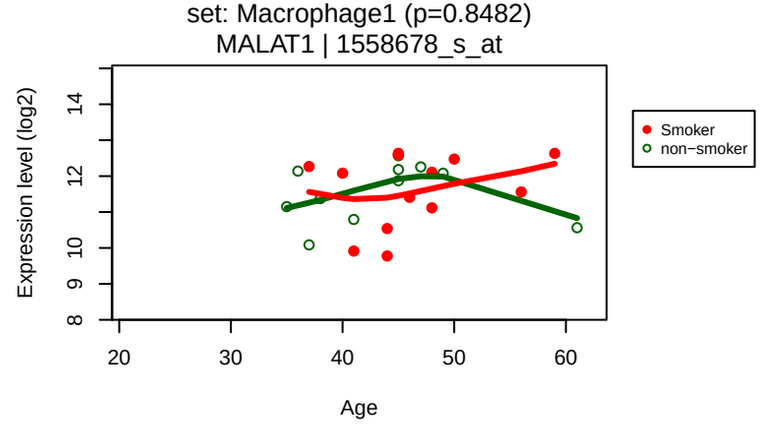

set: SmallAirway1 (p=0.5288)
MALAT1 | 1558678_s_at
Expression level (log2)
Age
Smoker
non−smoker

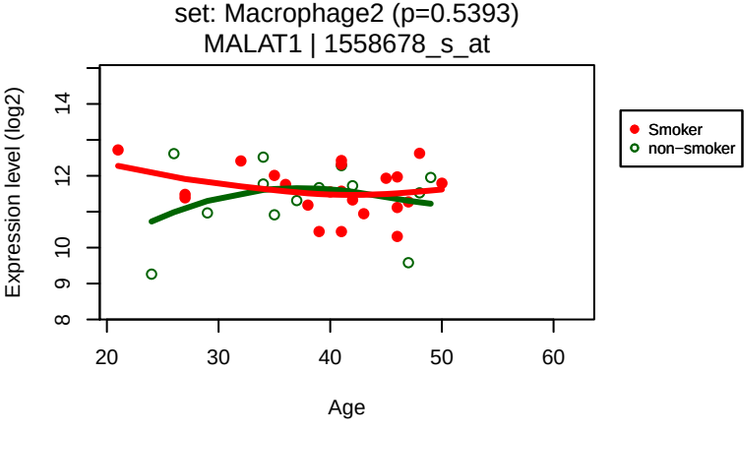

set: SmallAirway2 (p=0.035)
MALAT1 | 1558678_s_at
Expression level (log2)
Age
Smoker
non−smoker

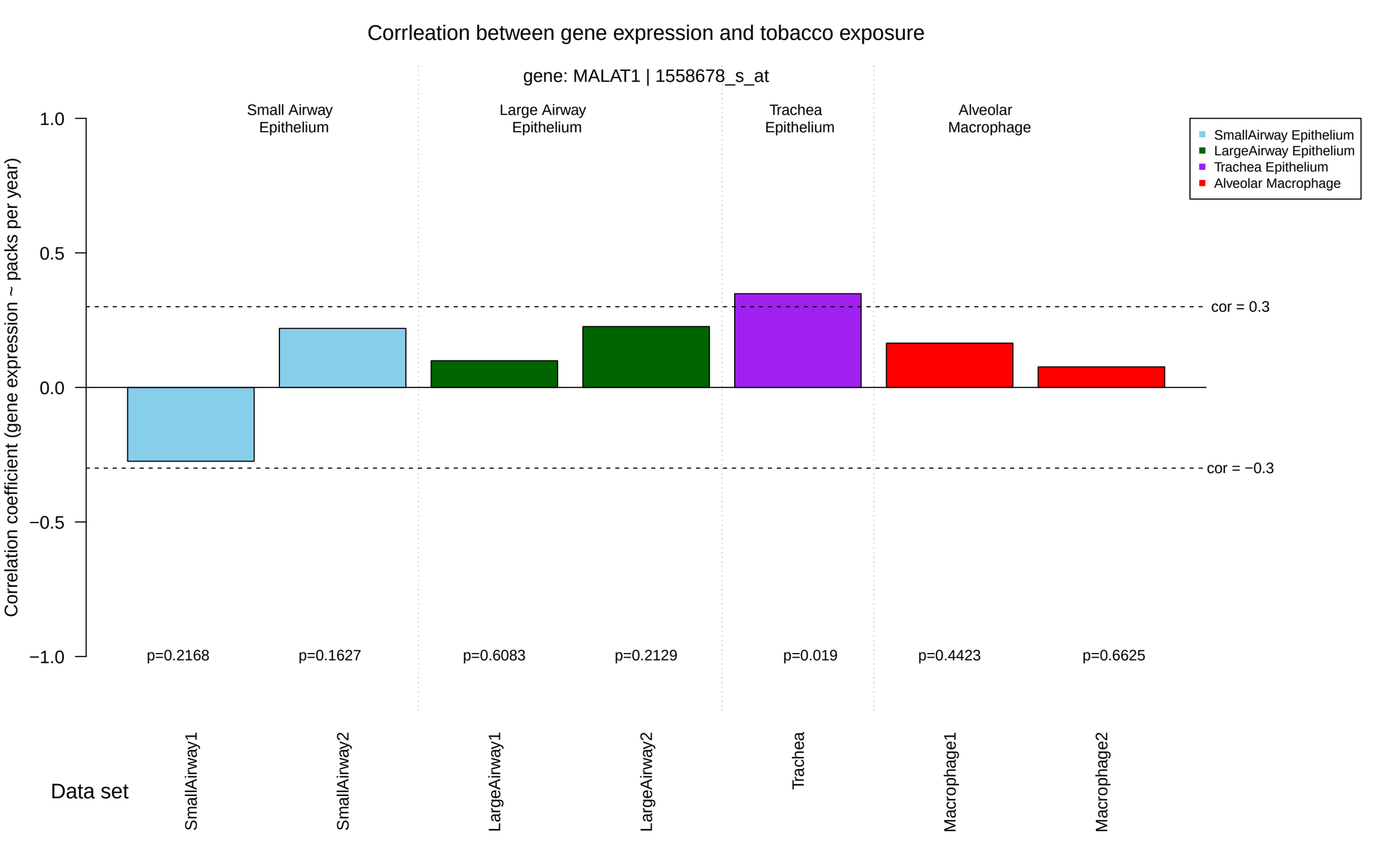

set: LargeAirway1 (p=0.8484)
MALAT1 | 1558678_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2233)
MALAT1 | 1558678_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0139)
MALAT1 | 1558678_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8482)
MALAT1 | 1558678_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5393)
MALAT1 | 1558678_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MALAT1 | 1558678_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2168
p=0.1627
p=0.6083
p=0.2129
p=0.019
p=0.4423
p=0.6625
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2